

WEEK 6 PROJECT SUMMARY

Data Analytics Track | Advanced Analytics & Decision Support

WEEK 6 AT A GLANCE

The Data Analytics track progressed HealthConnect from Week 5 exploratory analysis to a decision-oriented no-show risk analysis, improving the Power BI dashboard and translating observed patterns into business actions while documenting limitations and Week 7 testing needs.

1. Planned Accomplishment

Deepen the no-show analysis, validate important Week 5 findings, refine the dashboard, identify evidence-based interventions and prepare analytical inputs for the wider HealthConnect project.

2. What Was Completed

Four advanced analyses were completed covering appointment type, previous no-shows, booking lead time and reminder channel. The overall no-show KPI of 48.46% (2,423 of 5,000 appointments) was retained. The Power BI Week 6 page was redesigned with four decision-focused visuals, a Key Takeaway and Business Insight/Action panels. Limitations, recommendations and Week 7 testing requirements were documented.

3. What Improved From Week 5

The work moved from broad exploratory analysis to focused no-show-rate comparisons and business interpretation. The dashboard was also improved for clearer presentation through consistent sizing, a restrained color palette, a prominent KPI and concise action-oriented commentary.

4. What Was Integrated

The analysis was structured as a Data Analytics contribution to the wider HealthConnect workflow. Booking lead time and previous no-show history were identified as candidate inputs for subsequent predictive modelling and validation.

5. Cross-Track Collaboration

Collaboration with the other track was attempted, but no response or analytical output was received before submission. No unverified model changes, exchanged datasets or joint findings are therefore claimed.

6. What Was Exchanged and What Changed

No substantive cross-track exchange was completed before submission. As a result, there was no documented change to the Data Analytics work arising from received model outputs. The completed analysis provides a clear hand-off for future modelling, particularly around previous no-show history and booking lead time.

7. Key Findings / Development Outcomes

Outcome	Week 6 result
Strongest observed indicators	Longer booking lead times and previous no-show history show the clearest observed association with non-attendance.
Appointment type	No-show rates are relatively close across analysed appointment types, so appointment type alone is not a strong targeting basis.
Reminder channel	SMS has the lowest observed no-show rate among the analysed channels, but this requires further testing before causal conclusions are made.
Decision-support outcome	The dashboard now communicates both what the data shows and what HealthConnect could test or prioritise operationally.

8. Major Challenges

- Limited cross-track responsiveness before the submission deadline.
- Small sample sizes in some high previous-no-show categories required cautious interpretation.
- Reminder-channel differences are observational and cannot establish causation.
- Balancing analytical detail with a readable, business-presentation-ready dashboard.

9. Important Decisions

- Prioritised booking lead time and previous no-show history as the strongest observed risk indicators.
- Did not allow very small groups to drive the main conclusion.
- Presented reminder-channel findings as a testing opportunity rather than a proven intervention.
- Focused the final dashboard on four high-value visuals rather than overcrowding the page.

10. Remaining Issues

- Cross-track analytical integration remains incomplete because no collaborator output was received before submission.
- Observed relationships require further statistical and/or controlled testing.
- Potential confounding factors have not yet been fully assessed.
- Predictive modelling and prospective intervention validation remain future activities.

11. Contribution to the Overall HealthConnect Project

The Data Analytics contribution converts the appointment dataset into actionable evidence for the wider HealthConnect project. It identifies where non-attendance risk appears most concentrated, provides candidate

variables for future predictive modelling, highlights interventions that can be tested and establishes limitations so future decisions are evidence-based.

12. Proposed Focus for Week 7

- Validate whether booking lead time remains significant on additional or new data.
- Assess whether previous no-show history improves risk prediction beyond other available variables.
- Test reminder-channel effectiveness using appropriate statistical or experimental methods.
- Investigate possible confounding between reminder channel, patient characteristics and appointment characteristics.
- Compare the analytical findings with Data Science model results when available.
- Validate proposed operational interventions before wider implementation.